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MCY 650
Reading Assignment
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1. A sniffer needs your card to run in something called promiscuous mode. You need a special library/driver to put the card into promiscuous mode. What is the library/driver you need for Windows and Linux respectively?
Windows: WinPcap
Linux: LibPcap

2. What is Carrier Sense Multiple Access/Collision Detection (CSMA/CD)?
CSMA/ CD is a network protocol used to manage how data packets are transmitted over a shared communication medium, such as an Ethernet network. In Ethernet, as long as your system is within the same collision domain, you also receive every message intended for any other system in the same domain.

3. List 3 protocols that are susceptible to sniffing. Explain why.
SMTP: Written to carry ASCII, there is no security built in, everything is seen in clear text and can be read by someone sniffing the wire.
FTP: Requires a User ID and Password to access file server, info is passed in clear text over the wire.
TFTP, SNMP, POP3, and HTTP 'Application layer protocols': Info readily available to capture traffic.

4. What is the purpose of ARP protocol?
ARP is used to translate IP Address to physical addresses, i.e, MAC Address

5. How to show the current ARP cache? How to delete the ARP cache?
Linux
arp -n: Show the current ARP cache
arp -d [IP ADDRESS]: Delete the ARP cache

6. What is gratuitous ARP?
Gratuitous ARP is a special packet that updates the ARP cache of other systems before they even ask for it. Its original intent when created was to allow updates for outdated information, which helps with things like IP conflicts, clustering, and other legitimate issues. If you were to, somehow, change the ARP cache on Machine A and alter the cached MAC address of Machine B to your system's MAC address, you would receive all communication Machine A intended to send to Machine B. You can also change the ARP entry for the default gateway on all systems in

your subnet to your own machine. Now you're getting all the messages everyone was trying to send out of the local network, often the Internet.

7. What is active sniffing and passive sniffing?
Passive sniffing: Plug in a sniffer, and without any other interactions on your part, start pulling data packets to view at your leisure. Your machine's NIC must be a part of the same collision domain as the targets to listen. Hubs make sniffing easier.
Active sniffing: The collision domain you are a part of is segmented from those you want to look into i.e., by switch, so it requires some additional work on your part to force network devices to allow data packet viewing.

8. What is MAC Flooding?
MAC Flooding is to send so many MAC addresses to a CAM table, thus it can not keep up, effectively making it a hub. The CAM table is finite. It can be filled up fairly quickly and entries begin rolling off the list- this will allow attackers to see every packet on the switch.

9. What is DHCP Starvation?
DHCP Starvation is a DoS attack that targets a network's DHCP server by exhausting the pool of available IP addresses. The goal is to prevent legitimate devices from obtaining IP addresses, disrupting network connectivity.

10. What is a honeypot?
A **Honeypot** is a decoy system set up to entice attackers. It is designed to draw attackers in so you can watch what they do, how they do, and where they do it from. Two important points regarding Honeypots are first, any and everything found on the decoy system is not to be trusted. Second, the location of the decoy system is important. It should be in a not too easy place to access. Never placed openly within the network to prevent launch padding. Preferably in a DMZ boxed within firewalls to prevent further access to the network.